OpenFOAM

PENG

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1 Introduction

Definition 1.1 (openfoam)**.** *https://openfoam.org/resources/*

OpenFOAM is a free, open source computational fluid dynamics software to solve complex fluid flows involving turbulence and solid dynamics.

Components of OpenFOAM:

- solvers: perform calculation in fluid
- utilities: config simulation, geometry, mesh, and process results

2 Tutorials

Definition 2.1 (environment)**.**

```
# tutorial directory
$FOAM_TUTORIALS

# user run directory
$FOAM_RUN
mkdir -p $FOAM_RUN
```

3 Applications and Libraries

4 cases

5 Mesh Generation and Conversion

Definition 5.1 (mesh)**.** *OpenFOAM always operates in a three dimensional Cartesian coordinate system and all geometries are generated in three dimensions.*

OpenFOAM solves cases in two dimensions by specifying `empty` *boundary condition on boundaries normal to the third dimension for which no solution is required.*

Definition 5.2 (blockMesh)**.** *blockMesh generates meshes from a* `blockMeshDict` *file located in* `system` *directory for a given case.*

```
// -*- C++ -*-
// Website: https://openfoam.org
// Version: openfoam-dev

FoamFile // sub-dictionary
{
    format   ascii
    class    dictionary
    location "system"
    object   blockMeshDict
}

units [mm]; // optional
// the generated mesh is written in the base units for the underlying unit system, i.e. meter

vertices
(
    (x1 y1 z1) // vertex 1
    ...
);

blocks
(
    hex (v1 ... v8)
    (cells_x cells_y cells_z)
    simpleGrading (g_x g_y g_z)
    ...
);

boundary
(
    name
    {
        type T;
        faces
        (
            (v1 ... v4)
            ...
        );
    }
);
```

6 Boundary and Initial Conditions

Definition 6.1 (initial field)**.**

```
// case starts at t = 0
// initial field data is stored in a 0 sub-directory
```

7 Models and Physical Properties

8 Post-processing

Definition 8.1 (paraview)**.**

```
touch foam.foam

// Camera Parallel Projection
```

A Tools

Definition A.1 (installation)**.** *https://openfoam.org/download/source/*

```
cd ~/OpenFOAM/
git clone git@github.com:OpenFOAM/OpenFOAM-dev.git --depth=1
git clone git@github.com:OpenFOAM/ThirdParty-dev.git --depth=1
# git fetch --depth=1 && git reset --hard origin

cd OpenFOAM-dev
source "$HOME/OpenFOAM/OpenFOAM-dev/etc/bashrc"

./Allwmake
```